

Research project

The systematic discovery of novel metabolic islands using massive pan-genomics

Eoin Connell^{1,*}, Dr Paul Sheridan²

¹ College of Science and Engineering, School of Mathematics, Statistics Applied Mathematics, University of Galway, Galway, Ireland

*To whom correspondence should be addressed.

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Abstract

Motivation: Previous research has outlined the clustering of genes on the basis of functional association in the case of microbe defence systems. Given this premise this research project aimed to establish the localisation of genes associated with a known metabolism. Utilising genome databases: Progenomes V3, EGGNOG 4.5, GTDB v220 and the 2020 release of NCBI's COG a master dataset was produced. Panlocal a python script was then developed to carry out searches for genes within this dataset using their functional annotation. Key genes in this search associated with lactate metabolism were lactate permease, lactate racemase and a lactate dehydrogenase. Genes established as not being involved with lactate metabolism acted as controls, examples in GroEL and DnaK.

Results: The Panlocal script successfully produced resulting figures for genes associated with lactate metabolism across a wide variety of species. The control genes as expected appeared in significantly lower numbers in comparison to those associated with lactate metabolism. As a script Panlocal is applicable to the identification of further genes beyond those associated with lactate metabolism.

Contact: e.connell2@universityofgalway.ie

Supplementary information: Supplementary data and instructions on producing the dataset used and the Panlocal script are available at the following Github repository: <https://github.com/eoinc138/Panlocal> *Bioinformatics* online.

1 Introduction

In a research paper published by Doron et al, the process of searching in defined windows surrounding genes of known involvement in defensive systems in bacteria yielded the discovery of multiple new anti-phage defence systems present across multiple species of Bacteria[1]. This research paper involved multiple steps in regards to identifying these new defensive systems. They began with an initial search surrounding the area of known defence genes across multiple bacterial genomes and clustered genes found here based on sequence homology. Their downstream workflow then involved experimental verification of detected systems to confirm their defensive ability[1]. Inspired by the success of their work this research project endeavoured to potentially identify novel metabolic islands based on their proximity to an already well-researched and established metabolic process, in this case the metabolism of lactate by microbes. While Doron et al utilised sequence homology as the means of clustering the genes they used, the scope of this project across a large breadth of species necessitated a different means of clustering genes. EGGNOG provides a means of annotating genes based on function and as such allowed for clustering of genes with respect to their function[2].

The databases used in the final script as part of this project are Progenomes v3, EGGNOG v4.5, COG and GTDB release 220[3][2][4][5]. Each feature overlapping aspects which allowed for the subsequent production of the master dataset used in this research project. EGGNOG annotates genes based on their known functional relationship, in addition to this it annotates genes at an ortholog level[2]. Orthologs are more likely to retain ancestral functions in comparison to paralogs which arise from duplication events[2]. Given that the aim of this project specifically concerns the identification of genes based on their functionality, EGGNOG's method of gene annotation suited the aims of this research project. EGGNOG also offered the ability to functionally annotate at different taxonomic levels. Taxonomic annotation can go from as low as the species level to the phylum level[2]. As this research project contains a functional dataset comprising Euryarchaeota, Bacteroidetes, Actinobacteria, Firmicutes and Proteobacteria, EGGNOG's ability to annotate genes on the basis of the last universal common ancestor (LUCA) is key. Having a gene functionally annotated at the LUCA level ensures its representation in each of the five phylums, four being from Bacteria and one from Archaea. As such any search based on a specific annotation will not exclude any particular phylum.

The Genome taxonomy database (GTDB) is a "phylogenetically consistent and rank-normalised" database which comprises approximately

254,090 bacterial and 4,316 archaeal genomes, themselves further classified into 45,555 and 2,339 bacterial and archaeal species clusters respectively[5]. GTDB's database is often updated with additional species and reclassifications of its present species. This research project utilizes GTDB release 220 in order to produce a visual representation of the results from its dataset. The data provided by GTDB allows for the representation of these results at any taxonomic level ranging from Phylum to Species level along with the counts of the number of results present at each level.

The Database which provides the majority of the data for this research project is Progenomes, currently available at version 3[3]. Progenomes' data offered multiple benefits to issues which would have arisen had the data been sourced from other databases. Firstly, Progenomes collects its data into representative genome sets for each distinct species. For example, where most databases would feature an over-representation of often sequenced species such as *Escherichia coli*, Progenomes makes available a single representative for this species. This research project utilises the most recent version of Progenomes, V3 which features 41,171 of these representative genomes[5]. When conducting searches on a dataset this ensures no species receives an over-representation in found results. Progenomes also allows for the solving of over-representation through providing a contig ID corresponding to each gene ID. When conducting subsequent searches through the dataset, filtering to ensure genes found in our search lay on the same contig as our "anchor" gene prevented their over-representation. Progenomes also incorporates data from the EGGNOG database for each gene within the reference genome list[5]. As such carrying out a search on the basis of an EGGNOG functional annotation was made possible by the data available at Progenomes. Additionally each reference genome incorporates a NCBI biosample identifier[6]. To note subsequently in the text of this document, genome will be used to refer to Biosample. Figures may refer to genomes as Biosamples. From this the GTDB database can be linked as they provide taxonomic data for approximately forty seven thousand bacterial and archaeal genomes. The combination of these three databases provided the groundwork for constructing a large pangenomic dataset with a number of representatives across a diverse range of phyla. In deciding on a known metabolic process from which to conduct a search across, lactate metabolism arose as functional system of interest due to the abundance of available literature on the subject[7][8][9]. Additionally, being a system with a large number of functional genes within allowed for a means by which to narrow down our number of results[7]. By initially searching in a window around a chosen anchor gene and saving the resulting genes within the window, a second search could then be carried within these windows with a second "anchor" gene to further narrow down potentially associated systems, saving resulting windows where the initial anchor gene and second anchor gene are both present. In the case of lactate metabolism the gene encoding lactate permease was of great interest as virtually all means of lactate metabolism required an initial uptake of lactate into the cell facilitated by this permease[10][7]. This search around COG1620, the EGGNOG annotation for a lactate permease was then further refined by a subsequent search with a lactate racemase within the search window. Lactate racemase is key in facilitating further processing of lactate by converting the lactic acid between its D- and L- enantiomers[11][12]. As such the well understood mechanics underpin the means by which to establish an initial anchor search and a more refined search to narrow down an initial large number of results across the breadth of our dataset.

With these initial concepts established this research project endeavoured to create Panlocal, a gene search script written in Python. Its initial results in lactate metabolism as part of this research project are hoped to lay the groundwork for subsequent development of the scripts' functionality and application to genes in other biological systems.

2 Approach

Assemble a master dataset comprising a representative species for as many Bacterial and Archaeal phyla as possible. With this construct a computational script which searches for matching genes in each genome based on their functional annotation. Once found, the neighbouring genes in a defined window surrounding it are to be saved and written to a separate file for analysis. Further narrowing of results is to be carried out by considering a second functionally annotated gene. Produce an analysis of the most common genes within these search windows. Additionally, identify species where these gene clusters are present.

3 Methods

3.1 Initial dataset assembly

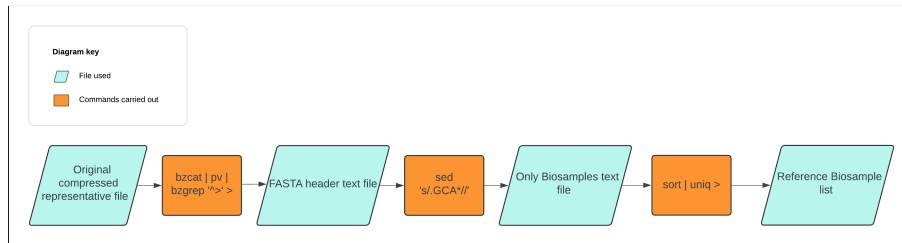


Figure 1. Process by which the reference protein file from ProgenomesV3 was converted to a text file comprising the reference Genome list. The initial file was ran through a decompression and parsing step using bzcat and bzgrep giving a FASTA headers file. This file was then subjected to a further parsing to remove protein sequence identifiers, sorted in descending order and removal of all duplicates. The final file being a text file of the Biosample ID for each species genome.

Assembly of the master dataset for this research project initially involved the creation of a reference genome list and reference gene list from the data available on Progenomes.

To begin creating a reference genome list the reference proteins file was downloaded from Progenomes[5]. As outlined in figure one this compressed file was then decompressed using a combination of bzcat and bzgrep to extract only the FASTA headers within which were subsequently written to a text file (Figure 1.). These FASTA headers were then subjected to further parsing by the sed command to remove everything from the .GCA characters onward. A combination then of sort and uniq linux commands removed all duplicates leaving a final reference list of the 41,171 genomes within (Figure 1.). For subsequent downstream parsing of large gene annotation files from Progenomes a reference gene ID list was required. This is outlined in figure 2 below.

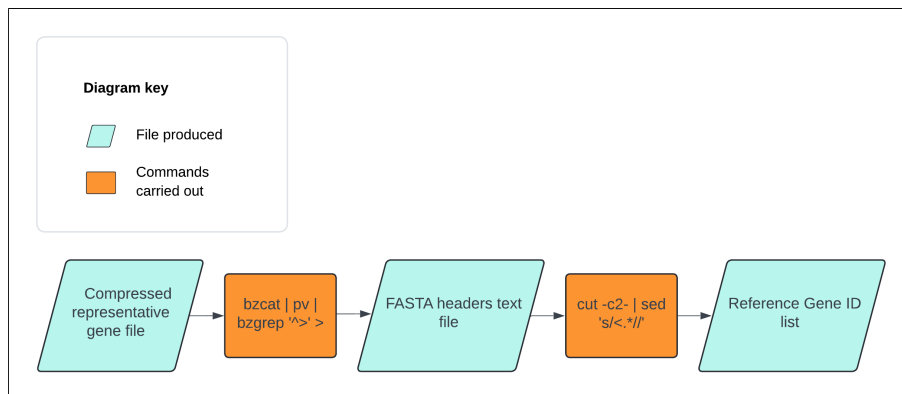


Figure 2. Process by which a reference gene file from ProgenomesV3 was converted to a text file of reference gene IDs. Firstly it was decompressed and parsed to give the FASTA header for each gene. A combination of cut and sed removed the remaining characters to give just the Gene ID.

Here similarly the compressed reference gene file was obtained from Progenomes[5]. It was then ran through the same combination of bzcat and bzgrep to give a text file of just FASTA headers. This text file was ran through sed and then cut to remove everything following the < character and the starting > character respectively. The final file then comprised a complete list of all reference Gene IDs for later parsing. With the reference genome and gene list it was then possible to create each phylum's dataset.

Figure 3 displays the steps required for the production of a dataset file for each individual phylum. It began with the download of each phylum's EGGNOG annotation file available from ProgenomesV3[5]. Using our reference genome list this file was parsed using an awk command to only include entries corresponding to each reference genome. Cut removed from this file any unnecessary columns leaving only the Biosample ID, Gene ID, EGGNOG annotations/ID etc. The separate Gene annotation files were then downloaded from Progenomes. This was cut then to leave only the two require columns, the gene ID and its corresponding contig ID. To note in the case of incredibly large phyla such as Proteobacteria, an additional round of parsing was required to carry out the subsequent workflow. This involved the reference gene list text file being used to parse the Gene annotations file to only include reference genes. This greatly reduced the file sizes involved and is noted in Figure 3 under the "Only for large phylums" banner. With a completed gene annotation file and dataset file derived from the EGGNOG annotation it was now possible to produce the finished dataset for the phylum. An awk command was used to match the contig ID from the gene annotation file to the dataset file based on the gene ID.

Figure 4 elaborates on the concatenation of each phylum. Awk was used to merge each phylum into a merged dataset. This was then sorted based on the first column to ensure each genome was together and that the genes within were in the correct order. This sorted dataset was then subject to a further awk command(*mistakenly excluded in figure 4) which removed the NCBI biosample ID from the first column and moved it to a new column at the end of the file. To create the header for the master dataset head and cut were used in conjunction to remove the first row from the Euryarchaeota EGGNOG

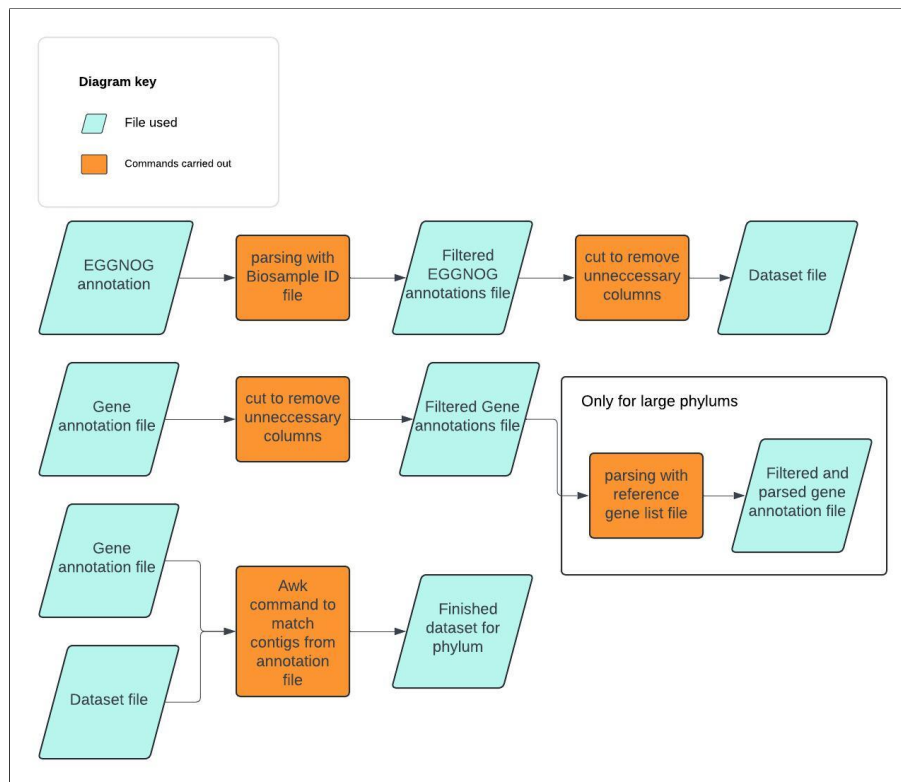


Figure 3. Process by which each Phylum's dataset was produced. Each phyla's EGGNOG annotation file was parsed to only contain reference biosamples, unnecessary columns were then removed giving a Dataset file. The gene annotation file was then matched against the Dataset file using awk to give the contig ID to each corresponding gene. In the case of large Gene annotation files, these were parsed using the reference gene file to only include reference dataset genes prior to matching with awk.

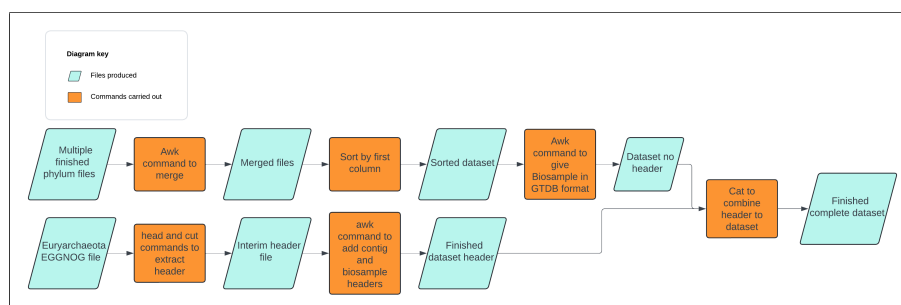


Figure 4. Process by which all phylum datasets were concatenated together. After merging via Awk they were sorted by biosample to ensure each species gene is in the correct order. Following this an awk command generated an additional column representing the Biosample in GTDB format to allow for matching of results with the GTDB taxonomy file. Simultaneously the header from the Euryarchaeota EGGNOG file is removed with additional columns added to designate contig ID and GTDB biosample entries. This is then concatenated using cat to complete the master dataset.

file. This was then given an additional two columns using awk to represent both the contig ID and NCBI biosample column. This final header was then added to the dataset using cat to give the final complete dataset for subsequent analysis. Given the incredibly large size of the final dataset of 37.9gb it was desirable to produce a smaller dataset from this to carry out initial quick test runs of the search script. Note that due to constraints on how the data for reference genomes is made available on ProgenomesV3 the final dataset did not comprise the total 41,171 reference genomes and 143,344,628 genes. Only 33,221 are represented with 106,288,259 genes therein.

As shown in figure 5 below, the first column of the master dataset was removed using cut and the ran through uniq to remove duplicate entries. Awk, wc-l and shuf commands were then used in conjunction to give a random sampling text file of five percent of the original text file. This file was then parsed by an awk command against the master dataset to produce a small test dataset representing five percent of the overall dataset. This process could be repeated any number of times to produce different test datasets. Additionally this process could also be followed against the dataset for each phylum to produce phylum test datasets if desired. Testing of the Panlocal script took place on Maxwell, a Linux super-computing cluster at the University of Aberdeen

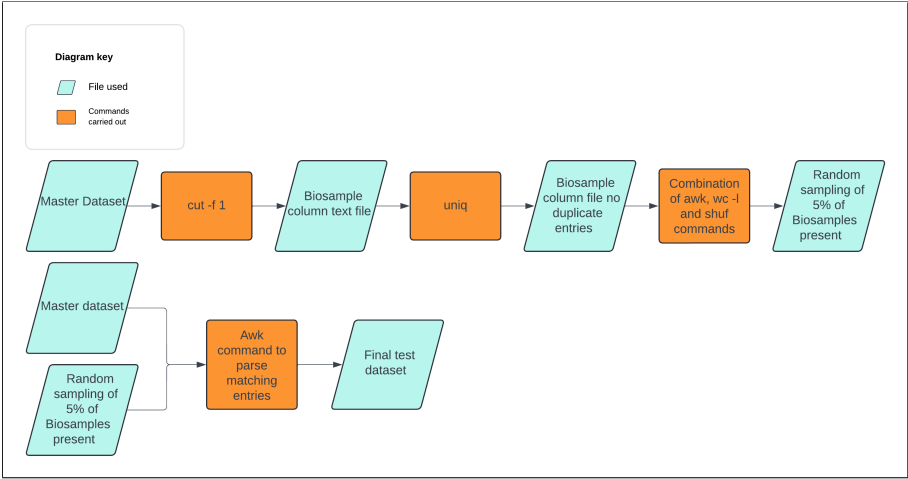


Figure. 5. Process by which a test dataset to assess Panlocal functionality was produced. Initially the Biosample ID column was parsed from the master dataset. Uniq used in conjunction with awk, wc -l and shuf removed any duplicate entries and obtained a random five percent of the total number of biosamples. This file of random biosamples was then parsed against the master dataset to produce a random test dataset five percent of the size of the total master dataset.

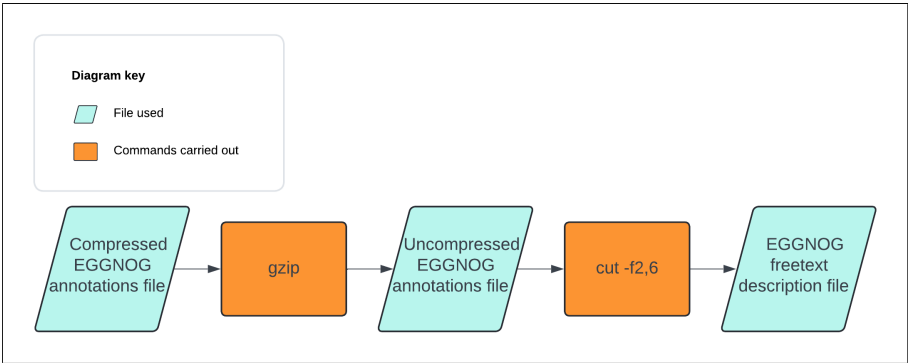


Figure. 6. Evolution of EGGNOG 4.5 annotations file to an EGGNOG descriptions tsv file. Initially the file was decompressed using gzip, it was then subsequently cut to remove unnecessary columns.

As part of the later search script a tsv file containing EGGNOG IDs, the counts for each of them and their description was to be produced. With how the script developed the most convenient way of producing this tsv file involved running the EGGNOG IDs from the search against another tsv file comprising the EGGNOG IDs and descriptions. The creation of this tsv file is outlined in figure 6. Initially the EGGNOG ID was obtained from the EGGNOG database. It was then decompressed through gzip and parsed to only include the EGGNOG ID and description column. Having the description/EGGNOG ID file separate also allows for incorporating new versions of the EGGNOG database into the search results more conveniently by following the procedure outlined in figure 6 on any new release. This research project aimed to remain consistent in databases used and as such used EGGNOG 4.5, as Progenomes V3 annotates using EGGNOG 4.5[5].

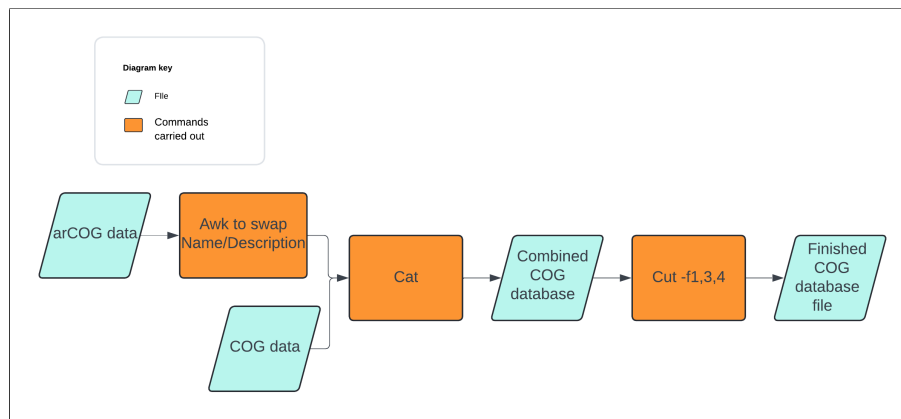


Figure 7. Evolution of COG database files. Initially both files were concatenated to give a combined file. Unnecessary columns were then removed using cut to give a final tsv file comprised of the COG/arCOG ID, gene name and description. Note: arCOG data is provided with Gene name and description columns swapped compared to COG data.

Although the EGGNOG annotation file provided descriptions of each Gene's EGGNOG entry it lacked great detail in terms of a Gene name. To account for this COG and arCOG's most recent database were referred to in order to obtain up to date gene names and descriptions corresponding to the EGGNOG IDs. The creation of a tsv file containing this information is outlined in Figure 7. As COG and arCOG IDs are represented in the EGGNOG database it was straightforward to integrate these annotations into the Panlocal workflow.

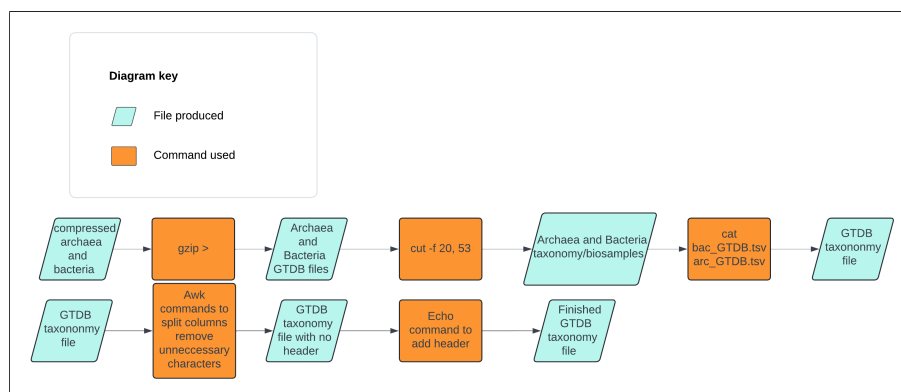


Figure 8. Production of a GTDB taxonomy tsv file. Initially taxonomic data for Archaea and Bacteria was obtained from GTDB v220. This was unzipped and cut to produce files containing our relevant columns: Biosample and taxonomic classification. These two files were then concatenated together. Using awk the taxonomy column was then split into multiple columns corresponding to each taxonomic levelCOG. Echo lastly added a header to complete the GTDB taxonomy file.

Assembly of a tsv file containing the taxonomy information for construction of our tree visualisation is elaborated on in figure 8. From the latest GTDB release the compressed data for Archaea and Bacteria was obtained. Once decompressed these files were then cut for their relevant columns, these being the NCBI biosample and GTDB taxonomic levels. The separate Archaea and Bacterial files were then combined using cat to give a complete taxonomic file. Further processing was carried out using awk to split the taxonomic column into individual columns for each taxonomic level. The characters denoting level were then removed from each entry in these new columns. Finally, echo was used to add a header denoting the corresponding NCBI biosample and each corresponding taxonomic level. In a similar fashion to the EGGNOG description file, having the GTDB taxonomic level file as a separate entity allows for ease of updating the script as GTDB updates their own database. Rather than annotation the master dataset with taxonomic information for each level, unnecessary bloating of the dataset can be avoided.

3.2 Panlocal script overview

With the prerequisite files produced, the python script for searching in predefined windows was written using these files to produce results. Figure 9 outlines the workflow of the python script below.

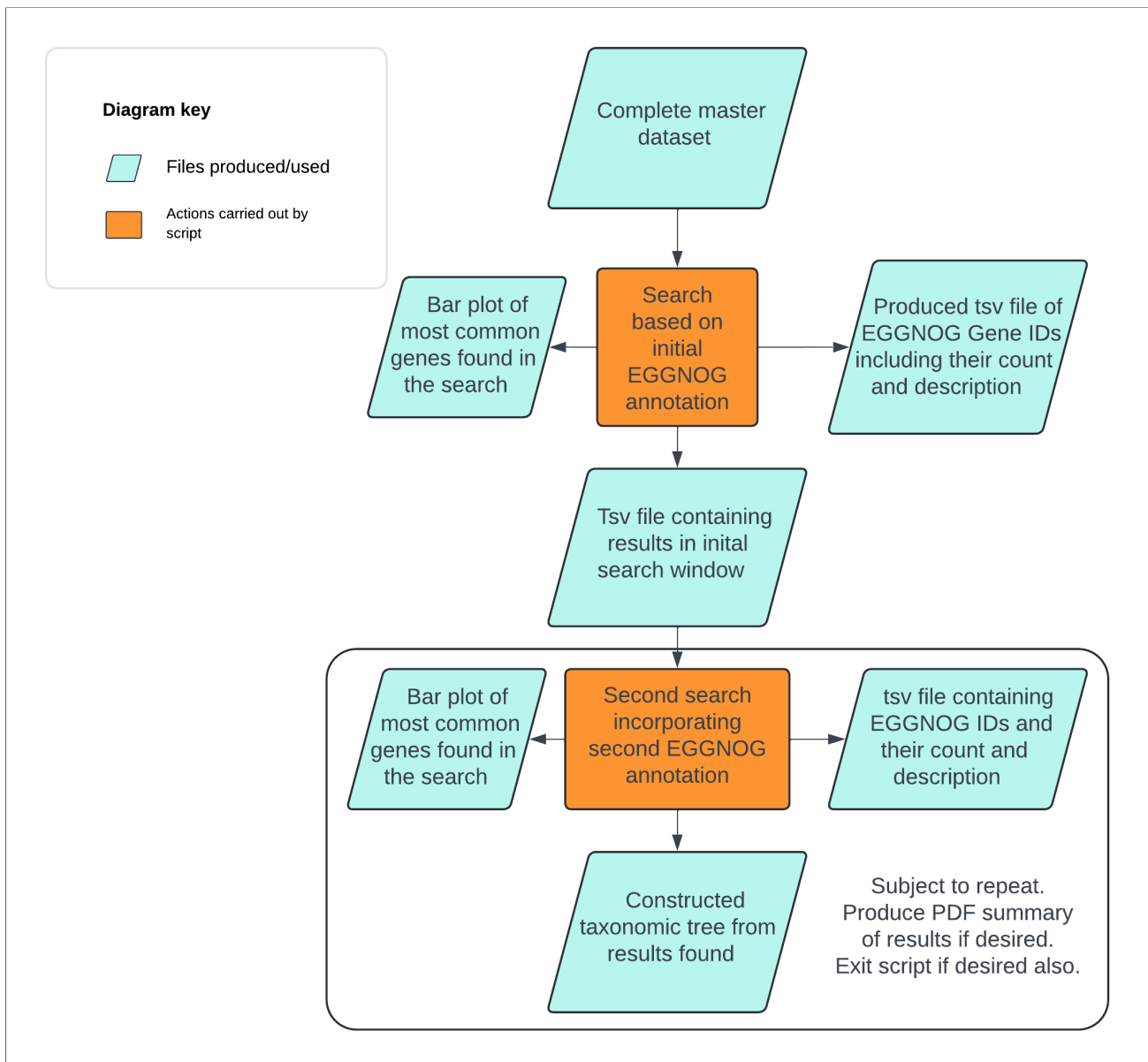


Figure. 9. Workflow of Panlocal script. Using the master dataset, an initial query EGGNOG annotation as well as search window is defined by the user. Genes lying in the window surrounding the query annotation are saved to a tsv file. A bar plot of the most common genes is produced as well as a tsv file of the most common EGGNOG IDs found in the window. At this point the user is prompted to produce a refined search using a second EGGNOG ID and a new search window. These inputs are searched in the initial search results tsv file parsing only genes lying around the original gene search which have the refined gene also present. The results are used to produce both a bar plot and EGGNOG ID tsv file again. Additionally a taxonomic tree can be produced of the species found from the refined search. At this point the user can request a PDF summary to be produced. After this the refined search can be repeated with a new EGGNOG ID or the script can exit depending on the users response.

Initially the user is queried on the EGGNOG ID for which they wish to search alongside a search window which spans the same distance both up and down the genome from this queried ID. Once the user provides the input the TSV file is firstly parsed for the Biosample column. With this Biosample column the counts for each genome are obtained. These counts are written to a temporary text file. Given the large size of the master dataset it must be broken up to allow for efficient processing on lower powered machines. As such the master dataset is broken up into chunks which are then searched individually. Each chunk's size corresponds to each genome ensuring that the search window around each gene does not overlap into a different genome in the event that the user's queried anchor gene is present at the beginning or end of a genome. The chunk size is set by the temporary text file derived from the biosample column. Once a chunk is read in to the script any matching genes are appended to a local list. Prior to appending genes are checked to ensure that firstly they do not lie on a different contig from that of anchor/queried gene and secondly that they have not already been appended to the master list comprising all results found from each genome. This ensures a two stage check of gene results. No gene can be appended if it does not lie on the same contig as the queried gene and if it has not already been saved to the overall results list.

Once the search through the master dataset has concluded the results are appended to a tsv file for later reference. A dynamic naming system ensures ease of retrieval from the users directory. This tsv file is then read in to the script again and ran through a counter on the basis of EGGNOG IDs specifically with the "@1" identifier. "@1" designate EGGNOG genes annotated on the last universal common ancestor basis[2]. Given that this research project spans species across both Bacteria and Archaea it is important that only genes annotated at this taxonomic level are considered. Any lower and our results would be limited to either Bacteria or Archaea and not meet the requirements of this project. The count of EGGNOG ID's is then plotted to show the top eight resulting EGGNOG IDs from the initial search. This file of EGGNOG IDs is then used to produce a tsv file containing four columns: their ID, Count, Gene name and Description. Firstly the IDs are parsed against the COG database file with matches being appended to a list. Following this they are parsed against the EGGNOG description file. Matches not already present in the list are given annotation information from the EGGNOG file. This two step method ensures that the resulting EGGNOG IDs are annotated firstly with the more detailed COG database but still receive descriptions where they are only present in the EGGNOG database. The user is also prompted to query specific Gene names, IDs, Descriptive keywords etc in this tsv file. Results are stored in another dataframe for later use.

At this point the script enters a checkpoint from which it can reset to. The user is then queried on a second EGGNOG ID to search for and define a new search window. Note that the window can not be expanded to be larger than the initial search window as it is searching in a file comprised of only the results from the first search. It can be reduced however to narrow the results produced. As the file being read is significantly smaller than the master dataset it no longer needs to be read in chunks. However the contig requirement still remains for appending genes. As such the genes returned can still only be recorded once. From the results of the second search the same bar plot and tsv file are produced as in the initial search result. Additional features are incorporated at this stage however. Biosample IDs in the tsv file are taken and have their duplicates removed. The resulting text file is then parsed against the GTDB taxonomy file with matching entries stored in a tsv file. Removing duplicates ensures that genomes with multiple loci are accounted for and are not represented more than once. The tsv file is used then to produce a tree visualisation of each taxonomic level represented in the results. At each taxonomic level the number of entries contained there in is displayed. The taxonomic tree can be remade at each taxonomic level: species, order etc if the user desires. The user is

then prompted on whether to produce a PDF summary file. Examples are included in the supplementary materials of this document and the Github repository. If accepted this produces a summary pdf containing the top 8 EGGNOG IDs from the initial search in a table, their bar plot representation and the collection of queried genes in a tabled format also. The refined search results are also saved in the same three means. This summary file is dynamically named depending on the initial and refined search gene IDs. After this stage the user is prompted on whether or not to repeat the refined search. Accepting restarts the script at the checkpoint but saves the outputs of the previous refined search. Declining exits the script with a concluding message.

3.3 Relevant genes to search and controls

Deciding on lactate metabolising genes using the Panlocal script primarily centered on those highlighted in a paper by Sheridan et al[7]. Here lactate permease, lactate racemase and FAD dependent D-lactate dehydrogenase stood out as candidate genes based on their representation across a variety of species. Comparison between the Racemase and Permease would also be of interest. Below in table 1 their names and corresponding EGGNOG IDs are collected. Both lactate permease and racemase served as initial anchor genes around which FAD dependent D-lactate dehydrogenase served as a secondary refinement gene. In both the initial and refined searches a search window of five was used. Establishing controls for this research

Table 1. Candidate Lactate metabolising genes

Gene name	EGGNOG ID
Lactate permease	COG1620
Lactate racemase	COG3875
FAD dependent D-lactate dehydrogenase	COG0277

project involved selecting genes expected not to cluster around our lactate metabolising genes. Given that the hypothesis of gene clustering based on function established by Doron et al we anticipated a lower number for these genes in comparison to our candidate genes. Control genes selected comprise a number of examples of genes encoding proteins heavily involved in DNA replication and protein folding. Additionally these genes are expected to associate with each other due to their relation in function. GroeL and GroeS alongside DnaK and DnaJ are the primary examples of this. Genes coding for these proteins were expected to be encountered across a wide variety of species in our dataset. Table 2 below collects our selected control genes. Each of these was queried in both the initial and refined search windows.

Table 2. Candidate control genes

Gene name	EGGNOG ID
GroeL	COG0459
GroeS	COG0234
DnaK	COG0443
DnaJ	COG0484
Gyrase B	COG0187
Ribosomal Large subunit L1	COG0081
Ribosomal Small subunit S4	COG0522

4 Results

4.1 Lactate permease search results

Table 3 below features the top 8 gene present in the Initial search window surrounding Lactate permease. To note is the presence of Lactate utilising genes, LutC and LutB. LldD and GlcD, 2 l-lactate dehydrogenases are present in significant number also.

Table 3. Initial search results for Lactate permease at a search window of five, L-lactate permease serves as the anchor gene.

EGGNOG ID	Count	Gene name	Description
COG1620	10719	LldP	L-lactate permease
COG0247	4304	GlpC	Fe-S cluster-containing oxidoreductase, includes glycolate oxidase subunit GlcF
COG0277	2702	GlcD	FAD/FMN-containing lactate dehydrogenase/glycolate oxidase
COG2186	2512	FadR	DNA-binding transcriptional regulator, FadR family
COG1556	2136	LutC	L-lactate utilization protein LutC, contains LUD domain
COG1139	1841	LutB	L-lactate utilization protein LutB, contains a ferredoxin-type domain
COG1304	1121	LldD	FMN-dependent dehydrogenase, includes L-lactate dehydrogenase and type II isopentenyl diphosphate isomerase
COG0583	1064	LysR	DNA-binding transcriptional regulator, LysR family

Table 4. Control genes in Initial search result for Lactate permease at a search window of five

EGGNOG ID	Count	Gene name	Description
COG0459	152	GroEL	Chaperonin GroEL (HSP60 family)
COG0234	138	GroES	Co-chaperonin GroES (HSP10)
COG0484	87	DnaJ	DnaJ-class molecular chaperone with C-terminal Zn finger domain
COG0443	54	DnaK	Molecular chaperone DnaK (HSP70)
COG0187	42	GyrB	DNA gyrase/topoisomerase IV, subunit B
COG0522	6	RpsD	Ribosomal protein S4 or related protein
COG0081	3	RplA	Ribosomal protein L1

Table 4 features each of the 7 control genes present although in significantly less numbers. GroEL and GroES are the most represented followed by DnaK and DnaJ. The remainder are severely under represented. Figure 10 shows the bar plot of the data from table 4, the near uniform representation of the remaining genes aside from Lactate permease is visible.

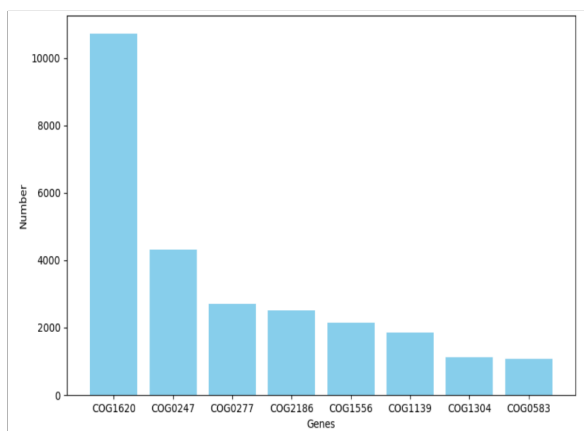


Figure 10. Bar plot of top eight most common genes within the initial search window for L-lactate permease(COG1620).

Table 5 shows the refined search results where only genes in a search window surrounding lactate permease that also contains FAD dependent D-lactate dehydrogenase are represented. GlpC and GlcD, both lactate dehydrogenases are represented in almost the same amount. The remaining genes in the window are almost all involved in either Lactate utilization or DNA-binding.

Table 5. Refined search results for Lactate permease as initial gene and FAD dependent D-Lactate dehydrogenase as refined search gene at a search window of five

EGGNOG ID	Count	Gene name	Description
COG1620	3103	LldP	L-lactate permease
COG0247	2901	GlpC	Fe-S cluster-containing oxidoreductase, includes glycolate oxidase subunit GlcF
COG0277	2702	GlcD	FAD/FMN-containing lactate dehydrogenase/glycolate oxidase
COG2186	1253	FadR	DNA-binding transcriptional regulator, FadR family
COG1556	837	LutC	L-lactate utilization protein LutC, contains LUD domain
COG1139	596	LutB	L-lactate utilization protein LutB, contains a ferredoxin-type domain
COG1304	487	LldD	FMN-dependent dehydrogenase, includes L-lactate dehydrogenase and type II isopentenyl diphosphate isomerase
COG0583	380	LysR	DNA-binding transcriptional regulator, LysR family

Figure 11 showcases the bar plot of data present in table 6. The near equal representation of Lactate permease and two D-lactate dehydrogenases is apparent.

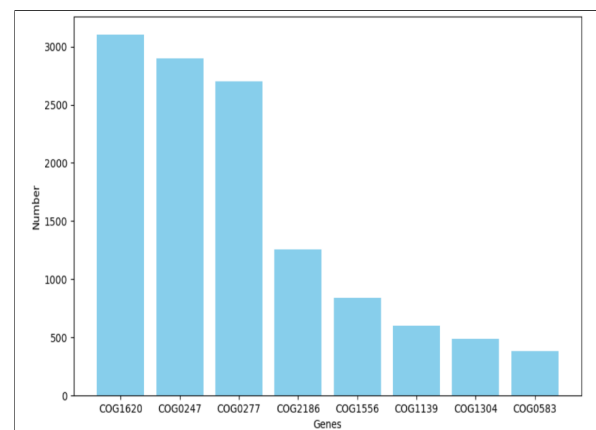


Figure 11. Top eight most common genes within the refined search window for L-lactate permease(COG1620) as initial gene and Lactate dehydrogenase(COG0277) as refined search gene.

Table 6 shows the control genes present in the refined search with Lactate permease and FAD-dependent D-lactate dehydrogenase. Gyrase B and the large Ribosomal subunit are noticeably not present at all. The remainder are present in equal volumes relevant to the genes they would be most associated with.

Table 6. Control genes in refined search result for Lactate permease and FAD-dependent D-Lactate dehydrogenase at a search window of five

EGGNOG ID	Count	Gene name	Description
COG0459	121	GroEL	Chaperonin GroEL (HSP60 family)
COG0234	117	GroES	Co-chaperonin GroES (HSP10)
COG0484	30	DnaJ	DnaJ-class molecular chaperone with C-terminal Zn finger domain
COG0443	24	DnaK	Molecular chaperone DnaK (HSP70)
COG0522	1	RpsD	Ribosomal protein S4 or related protein

Figure 12 contains the tree visualisation of the taxonomic data for the species represented in the refined search for Lactate permease. Notably Halobacteriota and Pseudomonata are the most represented for Archaea and Bacteria respectively.

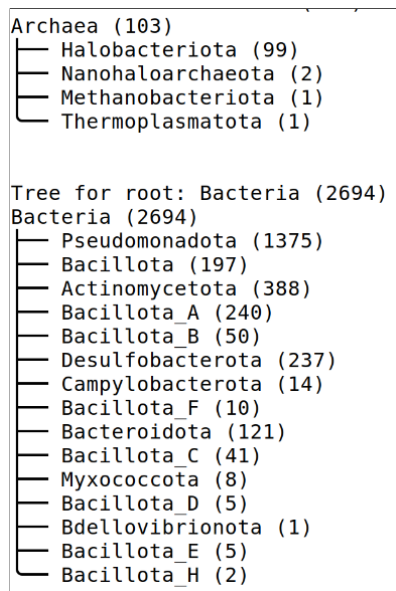


Figure 12. Tree visualisation of the refined results for Lactate permease(initial gene search) and FAD-dependent D-Lactate dehydrogenase(refined gene search), made up to the phylum level

4.2 Lactate racemase search results

Table 7 contains the top 8 genes found in the initial search window surrounding lactate racemase. Note that EGGNOG IDs: COG1691 and COG1606 lack a gene name due to a lack of one being present in either the COG or EGGNOG databases used. The significant decrease in the number of nearby genes is noticeable as well as the near equal volume at which they are present.

Figure 13 is a bar plot of the data from table 8. The disparity between Lactate racemase and neighbouring genes is apparent as well as the consistent number at which they appear.

Table 8 contains the control genes used for the initial search surrounding lactate racemase. Notable is the absence of Gyrase B and

Table 7. Initial search results for Lactate racemase(COG3875) at a search window of five

EGGNOG ID	Count	Gene name	Description
COG3875	6254	LarA	Nickel-dependent lactate racemase
COG1641	684	LarC	CTP-dependent cyclometallase, nickel-pincer nucleotide (NPN) cofactor biosynthesis
COG1028	659	FabG	NAD(P)-dependent dehydrogenase, short-chain alcohol dehydrogenase family
COG1691	638	-	NCAIR mutase (PurE)-related protein
COG1606	628	-	ATP-utilizing enzyme, PP-loop superfamily
COG0277	616	GlcD	FAD/FMN-containing lactate dehydrogenase/glycolate oxidase
COG0362	551	Gnd	6-phosphogluconate dehydrogenase
COG0204	520	PlsC	1-acyl-sn-glycerol-3-phosphate acyltransferase

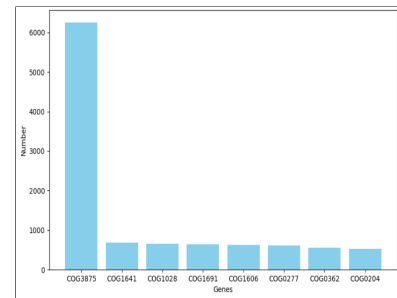


Figure 13. Top eight results from initial search for Lactate racemase(COG3875)

the large ribosomal subunit. Controls genes are noticeably present in a significantly lower amount compared to the lactate permease search.

Table 8. Control genes search results for Lactate racemase at a search window of five

EGGNOG ID	Count	Gene name	Description
COG0484	27	DnaJ	DnaJ-class molecular chaperone with C-terminal Zn finger domain
COG0443	12	DnaK	Molecular chaperone DnaK (HSP70)
COG0234	7	GroES	Co-chaperonin GroES (HSP10)
COG0459	5	GroEL	Chaperonin GroEL (HSP60 family)
COG0522	5	RpsD	Ribosomal protein S4 or related protein

Table 9 contains the refined search for Lactate racemase where D-lactate dehydrogenase acts as the refined search gene. Notable is the greater number of D-lactate dehydrogenases. Additionally, the presence of Lactate permease is noted.

Table 9. Refined search results for Lactate racemase(COG3875) as initial gene and FAD dependent D-lactate dehydrogenase(COG0277) as refined search gene at a search window of five

EGGNOG ID	Count	Gene name	Description
COG0277	616	GlcD	FAD/FMN-containing lactate dehydrogenase/glycolate oxidase
COG3875	610	LarA	Nickel-dependent lactate racemase
COG0247	281	GlpC	Fe-S cluster-containing oxidoreductase, includes glycolate oxidase subunit GlcF
COG2086	230	FixA	Electron transfer flavoprotein, alpha and beta subunits
COG2025	211	FixB	Electron transfer flavoprotein, alpha subunit FixB
COG1620	174	LldP	L-lactate permease
COG2186	169	FadR	DNA-binding transcriptional regulator, FadR family
COG1960	69	CaiA	Acyl-CoA dehydrogenase related to the alkylation response protein AidB

Figure 14 features the data from table 10 represented in a bar plot. The near equal present of Lactate racemase and Lactate dehydrogenase is noticeable.

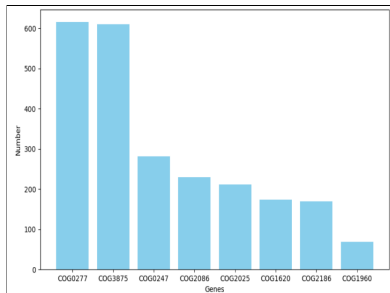


Figure. 14. Top eight most common genes within the refined search window for lactate racemase(COG3875) as initial gene and lactate dehydrogenase(COG0277) as refined search gene.

Table 10 contains the number of control genes present in the refined search with Lactate racemase and Lactate dehydrogenase. Only DnaJ, DnaK and the small ribosomal subunit are represented. Their extremely low counts is apparent.

Table 10. Control genes in refined search result for Lactate racemase(COG3875) and FAD-dependent D-Lactate dehydrogenase(COG0277) at a search window of five

EGGNOG ID	Count	Gene name	Description
COG0484	11	DnaJ	DnaJ-class molecular chaperone with C-terminal Zn finger domain
COG0522	2	RpsD	Ribosomal protein S4 or related protein
COG0443	1	DnaK	Molecular chaperone DnaK (HSP70)

Figure 15 is the tree visualisation of the refined search with Lactate racemase and Lactate dehydrogenase at the phylum level. Halobacteria is notable present as the main Archaeal phylum. Bacillota A and Desulfobacterota are the main Bacterial representatives.

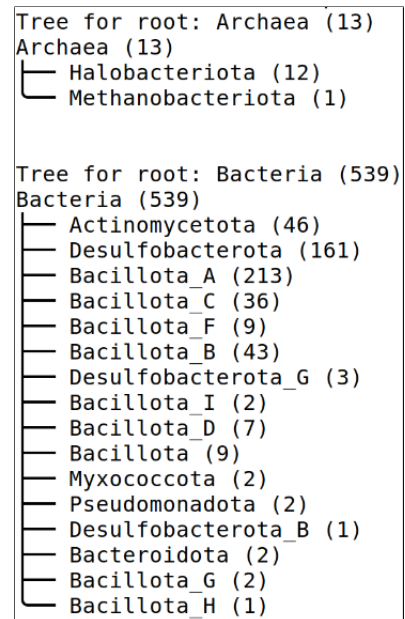


Figure. 15. Tree visualisation of the refined results for Lactate racemase(initial gene search) and FAD-dependent D-Lactate dehydrogenase(refined gene search), made up to the phylum level

5 Discussion

5.1 Lactate permease results

From the results shown it can be deduced that the Panlocal script successfully carried out its function of clustering genes with respect to their functional annotation. Lactate utilizing genes such as Lactate permease or Lactate racemase appear to lie in genomes surrounded by genes which take part in further lactate metabolism. The much greater number of results for lactate permease compared to lactate racemase can be attributed to the importance of lactate permease in allowing lactate to enter microbes for further conversion into pyruvate etc[7]. The genes in the initial search all correspond to differing aspects of lactate utilisation. COG0247 corresponding to GlpC is a noted gene involved in a lactate metabolic process from the EGGNOG. Note that COG0247 and COG0277 both annotate for lactate dehydrogenase activity in the EGGNOG database. Lactate dehydrogenase is a key factor in modes of lactate metabolism in different species[13]. FadR and LysR are noted transcriptional regulators[14][15]. LutC and LutB are noted to be involved in lactate oxidation[16]. LldD is another gene involved in lactate metabolism as another lactate dehydrogenase, current literature highlights it necessity for growth on L-lactate[17]. In regards to the controls used for the initial search their numbers are much less in comparison. GroEL and GroES have close numbers as do DnaK and DnaJ, likely reflections of their own close functional relationship[18][19].

When a lactate dehydrogenase is used to refine the initial results it can be seen that the same trend repeats. The two nearest genes in count are GlpC and GlcD, implying a very close functional relationship between these and lactate permease. GlpC and GlcD are associated with organic solvent tolerance and glycolate oxidation respectively[20][21][22]. The same transcription factor genes are notably present along with the same LutB and LutC. Control numbers fall more significantly with Gyrase B and a ribosomal subunit gene not appearing. The same trend of closeness is observed for GroEL/GroES and DnaK/DnaJ. Interestingly the refined results appear predominantly in the Halobacteriota and Pseudomonadota phyla. For Halobacteriota this may reflect the metabolism of its member

species, however further research would be needed to assert this claim[23]. Pseudomonadota may contain the bulk of results as its members fall under Proteobacteria taxonomic classification, the phylum which makes up the bulk of our dataset.

5.2 Lactate racemase results

In table 8 the initial search using lactate racemase returns a wide variety of genes. FabG and GND are associated with encoding an malonyl coenzyme A-acyl carrier protein transacylase and dehydrogenase activity respectively[24][25]. GlcD is present again. LarC is present as well in the highest number after Lactate racemase. Lactate racemase is encoded by the Lar locus on which LarA, LarB, LarC, LarD and LarE are all located and required to allow for lactate racemase function[11]. Only when the search results are refined using the lactate dehydrogenase do more genes appear associated with lactate metabolism. Notable in table 10 is the clustering of genes associated with acting as lactate dehydrogenases in the vicinity of the lactate racemase. After the conversion of lactate between its enantiomers its is acted upon by a D-lactate hydrogenase[7]. As such given that a dehydrogenase follows the racemase in the metabolism pathway the results here align with expectations. The lack of genes associated with lactate utilisation based on the initial search for lactate racemase is apparent when compared with the initial search using lactate permease. It can be interpreted as implying lactate permeases more direct relationship with other lactate utilising genes. Additionally the results for lactate racemase can also be seen to highlight the effectiveness of using a second anchor gene. The results from the second search feature genes much more directly associated with lactate metabolism.

6 Future perspectives

While the Panlocal script clearly functions in delivering a resulting collection of genes surrounding a gene of interest it does suffer from several shortcomings. Most obvious of these is the missing representative genomes from Progenomes. While Progenomes does make EGGNOG annotation information available for the its genome entries is is not easily accessible for the reference genomes. The phylum level is the highest possible taxonomic classification at which bulk EGGNOG annotations are available, limiting our ability to match the reference genomes with EGGNOG annotation. Thankfully the modular nature in which the dataset was assembled may allow future versions of progenomes which incorporate EGGNOG annotation in their reference dataset to be easily integrated. Greater insights into the distribution of genes would be possible with the additional 7,950 genomes present. Further to the modular nature of the Panlocal script future updates for its composite files: EGGNOG, COG and GTDB can easily be integrated. As of the beginning of this research project GTDB for example was on its most recent release, v220 in April of this year. Assuming a further update in the near future, the new taxonomic classifications could easily be downloaded and integrated into the script. In regards to the computational potential of Panlocal the usage of a chunk system to divide the master dataset provides two separate avenues of potential. Firstly the dividing of the dataset severely reduces the computational power required. When running the script the initial search takes between 10 15 minutes depending on the query and hardware available. On a modern quad-core laptop the CPU usage never exceeded 15 percent while the ram usage was a modest 300mb approximately. As such Panlocal is able to be ran on a wide variety of hardware opening up its potential use to individuals of any background. Secondly the splitting of the gene search on the basis of genome allows for the potential application of hyper-threading to speed the scripts progress. On a high performance cluster, multiple cores could be applied to the running a search on each genome. Search times could greatly be improved and bulk production of

search data could be carried out in a reduced time frame. The successful use of Panlocal on Maxwell the high performance cluster at the University of Aberdeen showcases its adaptability to already run on a wide variety of hardware.

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